

# ngspice-46 vs. Spectre — Feature Gap Analysis

Ngspice + OpenVAF Enhancements project

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## ngspice-46 vs. a commercial simulator (Spectre) — feature gap analysis

A capability comparison of the ngspice-46 build in this repository against a Spectre-class commercial simulator, to show where the open-source tool already matches the commercial one and where the gaps are. Grounded in a direct read of the ngspice-46/src/ tree (analyses, solver, devices, RF, convergence, parallelism), not marketing feature lists.

Legend: ✓ present / on par · ⚠ partial, experimental, or via a workaround · ✗ absent. "Spectre" stands in for the commercial reference (Spectre / SpectreRF / Spectre X / RelXpert feature set).

The one column that matters is ngspice — Spectre has essentially everything, so its column is a baseline of ✓. The point of the table is where ngspice's mark is ⚠ or ✗.

Context: the Verilog-A / OSDI device side is not a gap — thanks to the OpenVAF-reloaded compiler in this repository it is on par with (often ahead of) commercial Verilog-A support. The gaps below are all on the simulator/analysis side.

### Core numerics

| Category    | Feature                                         | ngspice | Spectre |
|-------------|-------------------------------------------------|---------|---------|
| Core solver | Sparse LU (KLU) direct solver                   | ✓       | ✓       |
| Core solver | Legacy Sparse1.3 fallback                       | ✓       | ✓       |
| Core solver | Parallel / partitioned / GPU linear solve       | ✗       | ✓       |
| Core solver | Matrix reordering + scaling beyond KLU defaults | ⚠       | ✓       |
| Integration | Trapezoidal + variable-order Gear               | ✓       | ✓       |
| Integration | Advanced LTE-based step/order control           | ⚠       | ✓       |
| Convergence | gmin stepping + source stepping homotopy        | ✓       | ✓       |
| Convergence | Pseudo-transient / dynamic-gmin continuation    | ⚠       | ✓       |
| Convergence | Damped / trust-region (globalized) Newton       | ⚠       | ✓       |
| Convergence | Coordinated accuracy presets (errpreset)        | ✓       | ✓       |

The two convergence ⚠ rows: ngspice's DC path is not naked — it has per-device junction limiting (30 device families), an optional `nodedamping` step clamp, and a multi-stage homotopy cascade (`dynamic_gmin` → `new_gmin` → `spice3_gmin` → `source stepping`). What it lacks vs. commercial tools is a principled globalized Newton (residual-based trust-region / Armijo) and a classic pseudo-transient ( $\dot{X}$ -embedded) continuation. [Enhancement-111](#) adds the former: `.option linesearch`, an Armijo backtracking line search on a new KCL-residual merit  $F = \|G \cdot x - b\|$  (result-neutral, off by default).

### Standard analyses (analog)

| Category | Feature                                                  | ngspice | Spectre |
|----------|----------------------------------------------------------|---------|---------|
| Analyses | DC operating point / DC sweep                            | ✓       | ✓       |
| Analyses | Transient (.tran)                                        | ✓       | ✓       |
| Analyses | AC small-signal (.ac)                                    | ✓       | ✓       |
| Analyses | Noise, small-signal (.noise, onoise/inoise + integrated) | ✓       | ✓       |
| Analyses | Pole-zero (.pz)                                          | ✓       | ✓       |
| Analyses | Transfer function (.tf)                                  | ✓       | ✓       |
| Analyses | DC sensitivity (.sens)                                   | ✓       | ✓       |
| Analyses | Distortion (.disto)                                      | ✓       | ✓       |
| Analyses | Measurement / post-processing (.meas)                    | ✓       | ✓       |

## RF / periodic steady-state suite

| Category | Feature                                   | ngspice | Spectre |
|----------|-------------------------------------------|---------|---------|
| RF       | S-parameter analysis + noise figure (.sp) | ✓       | ✓       |
| RF       | Touchstone (.snp) import / export         | ✓       | ✓       |
| RF       | Periodic steady state (PSS)               | ⚠       | ✓       |
| RF       | Harmonic Balance (HB)                     | ×       | ✓       |
| RF       | Periodic / phase noise (Pnoise)           | ×       | ✓       |
| RF       | Periodic AC (PAC, conversion gain)        | ×       | ✓       |
| RF       | Periodic transfer function (PXF)          | ×       | ✓       |
| RF       | Periodic S-parameters (PSP)               | ×       | ✓       |
| RF       | Quasi-periodic / multi-tone (QPSS / QPAC) | ×       | ✓       |
| RF       | Envelope following                        | ×       | ✓       |

PSS is ⚠ present but a brute-force shooting method, flagged experimental, and fragile on some circuits. HB exists only as a WITH\_HB stub that returns "unsupported".

## Performance & scale

| Category    | Feature                                            | ngspice | Spectre |
|-------------|----------------------------------------------------|---------|---------|
| Performance | Multithreaded device-model evaluation (OpenMP)     | ✓       | ✓       |
| Performance | Element bypass / latency exploitation              | ⚠       | ✓       |
| Performance | Fast-SPICE hierarchical / isomorphic engine        | ×       | ✓       |
| Performance | Distributed (MPI) / cloud partitioning             | ×       | ✓       |
| Performance | Handles $10^6$ – $10^8$ -node post-layout netlists | ×       | ✓       |

## Statistical / variability / yield

| Category    | Feature                                                       | ngspice | Spectre |
|-------------|---------------------------------------------------------------|---------|---------|
| Statistical | Monte Carlo                                                   | ⚠       | ✓       |
| Statistical | Native process/mismatch modeling + correlations               | ⚠       | ✓       |
| Statistical | Low-discrepancy sampling (Sobol / Latin-hypercube)            | ×       | ✓       |
| Statistical | High-sigma methods (importance sampling, worst-case distance) | ×       | ✓       |
| Statistical | Corner + MC + yield estimation flow                           | ⚠       | ✓       |

Monte Carlo is ⚠ works, but script-driven (alter + sgauss, with the deterministic-seed RNG from Enhancement-10/66) rather than a native statistical block with sampling controls.

## Reliability / aging

| Category    | Feature                                          | ngspice | Spectre |
|-------------|--------------------------------------------------|---------|---------|
| Reliability | Device aging (HCI / NBTI / TDDB)                 | ×       | ✓       |
| Reliability | Stress → degrade → re-simulate (fresh/aged) flow | ×       | ✓       |
| Reliability | Electromigration + IR-drop (EMIR)                | ×       | ✓       |

## Post-layout / parasitics

| Category    | Feature                                | ngspice | Spectre |
|-------------|----------------------------------------|---------|---------|
| Post-layout | Flat parasitic (RC) netlist simulation | ✓       | ✓       |
| Post-layout | RC reduction / model-order reduction   | ×       | ✓       |
| Post-layout | n-port (S/Y/Z) extracted-block import  | ✓       | ✓       |

## Behavioral / mixed-signal / AMS

| Category | Feature                                    | ngspice | Spectre |
|----------|--------------------------------------------|---------|---------|
| Modeling | Verilog-A (analog, LRM Annex C) via OSDI   | ✓       | ✓       |
| Modeling | XSPICE code models + event-driven digital  | ✓       | ✓       |
| Modeling | Verilog-AMS mixed-signal (connect modules) | ×       | ✓       |
| Modeling | Real-number modeling (wreal / RNM)         | ×       | ✓       |
| Modeling | VHDL-AMS                                   | ×       | ✓       |

## Interfaces & infrastructure

| Category       | Feature                           | ngspice | Spectre |
|----------------|-----------------------------------|---------|---------|
| Interface      | Shared-library / programmatic API | ✓       | ✓       |
| Interface      | Python bindings                   | ✓       | ✓       |
| Infrastructure | Built-in optimizer                | ×       | ✓       |
| Infrastructure | Checkpoint / restart of long runs | ⚠       | ✓       |

## Where to invest (given the Verilog-A/OSDI side is done)

The realistic, winnable identity for this project is the open-source, OSDI/Verilog-A-native simulator with a real RF-noise and statistical story — not out-engineering 25 years of commercial fast-SPICE/parallel work. Ranked by leverage on the existing strength × differentiation × tractability:

1. RF periodic small-signal suite — Pnoise → PAC → PXF, on a hardened PSS. The device/OSDI side already supplies what these need (noise-source topology, operating-point- and frequency-dependent load\_noise, periodic Jacobians); the work is the analysis engines. Genuinely novel in open source.
2. Convergence robustness — coordinated accuracy presets (errpreset) landed in [Enhancement-110](#); the remaining piece is pseudo-transient / dynamic-gmin homotopy. Unglamorous but makes every analysis usable on real circuits; self-contained in the ngspice core.
3. High-sigma statistical sampling — high industrial value (yield / SRAM), moderate difficulty, leans on the deterministic-seed RNG already in place.

Explicitly lower priority: fast-SPICE parallelism and aging/EMIR — enormous efforts that don't leverage what makes this project distinctive.